Chittagong University of Engineering & Technology

Department of Electronics and Telecommunication Engineering

EXPERIMENT NO: 02

Name of the Experiment

Implementing Local Area Network Using Hub and Switch

Course No : ETE 452

Course Title : Advanced Communications and Networking Sessional

Date of Experiment : 11/01/2026

Date of Submission : 25/01/2026

REMARKS

Objectives

- To implement a LAN using a hub and a switch.
- To observe data communication among multiple PCs in a LAN.

Software

- Cisco Packet Tracer

Introduction

A Local Area Network (LAN) connects multiple devices within a limited area.
A hub broadcasts data to all ports, whereas a switch forwards data to specific devices.
LAN improves resource sharing and communication efficiency.

Procedure

1. Multiple PCs were added to the workspace.
2. A hub and a switch were placed separately for implementation.
3. Straight-through cables were used to connect PCs to the hub and switch.
4. IP addresses were assigned to all PCs in the same subnet.
5. Connectivity was verified using the ping command.

Connection Diagram

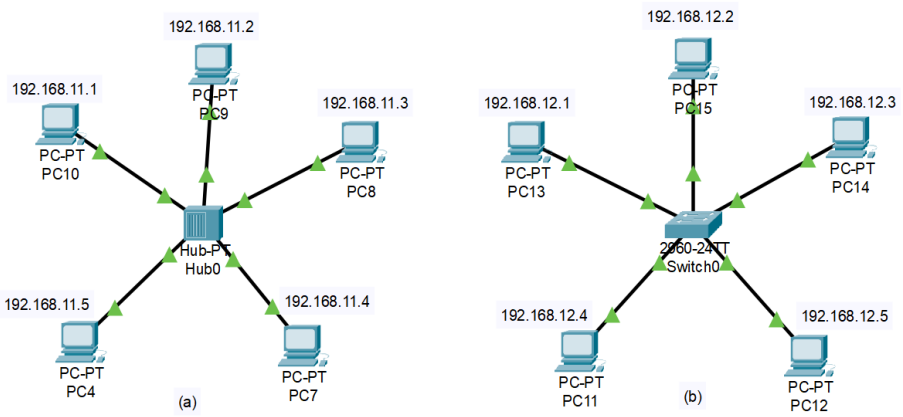

Figure 2 : Local Area Network (LAN) using (a) Hub and (b) Switch

A switch is considered smarter than a hub because MAC address learning and forwarding decisions are performed using a MAC address table. Frames are forwarded only to the specific destination port. In contrast, a hub does not perform MAC address resolution and blindly broadcasts data to all connected ports

Discussion

1. LAN connectivity was successfully implemented using both hub and switch.
2. It was observed that the hub broadcasted data to all connected PCs.
3. The switch transmitted data only to the intended destination PC.
4. Reduced network traffic was noticed in the switch-based LAN.
5. This experiment highlighted the performance difference between hub and switch.